Cambridge (CIE) A Level Computer Science

Your notes

Files

Contents

- File Handling

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Purpose of files

What is file handling?

- File handling is **the use of programming techniques** to work with information stored in **text files**
- Examples of file handing techniques are:
  - **opening** text files
  - **reading** text files
  - **writing** text files
  - **closing** text files

Concept	CIE A Level Pseudocode
Open file for reading	OPENFILE fruit.txt FOR READ
Open file for writing	OPENFILE Shopping.txt FOR WRITE
Close file	CLOSEFILE fruit.txt
Read line	READ fruit.txt, line
Write line	WRITE fruit.txt, "Oranges"
Check end of file	IF NOT EOF(fruit.txt) THEN
Create new file	OPENFILE Shopping.txt FOR WRITE (same as writing – it overwrites or creates)
Append to file	OPENFILE Shopping.txt FOR APPEND

- An example program written in pseudocode to:
  - **Opens** a text file for reading
  - **Reads** each line (a fruit name)
  - **Counts** how many fruits are in the file
  - **Outputs** the total

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- **Closes** the file

```
DECLARE FruitName : STRING
DECLARE FruitCount : INTEGER

FruitCount ← 0

OPENFILE FruitFile FOR READ

WHILE NOT EOF(FruitFile) DO
   READ FruitFile, FruitName
   FruitCount ← FruitCount + 1
ENDWHILE

CLOSEFILE FruitFile

OUTPUT "Total number of fruits: ", FruitCount
```

Identifier table

Identifier	Data type	Description
FruitName	STRING	Stores the name of each fruit read from the file
FruitCount	INTEGER	Keeps track of how many fruits have been read
FruitFile	FILE	The file containing the fruit names (e.g. "fruit.txt")

Worked Example

Write program code to read the contents of Data.txt into DataArray

Answer

- Opening file Data.txt to read [1 mark]
- Looping through all the 25/EOF ... [1 mark]
- ... reading each line and storing/appending into array [1 mark]
- Exception handling with appropriate output [1 mark]
- Closing the file (in an appropriate place) [1 mark]

Example program code:

Java

```
Integer Counter = 0;
try{
   Scanner Scanner1 = new Scanner(new File("Data.txt"));
   while(Scanner1.hasNextLine()){
      DataArray[Counter] = Integer.parseInt(Scanner1.next());
      Counter++;
   }
   Scanner1.close();
}catch(FileNotFoundException ex){
```

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```
    System.out.println("No data file found");
}
```

VB.NET

```
try
  Dim DataReader As New System.IO.StreamReader("Data.txt")
  Dim X As Integer = 0
  Do Until DataReader.EndOfStream
    DataArray(X) = DataReader.ReadLine()
    X = X + 1
  Loop
  DataReader.Close()
Catch ex As Exception
  Console.WriteLine("Invalid file")
End Try
```

Python

```
try:
  DataFile = open("Data.txt",'r')
  for Line in DataFile:
    DataArray.append(int(Line))
  DataFile.close()
except IOError:
  print("Could not find file")
```

Your notes

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